

M303 Book D Day School Questions - 07 February 2026 - by Joe Kyle

The “Extras” are more challenging exercises; something to look at **after** the TMA (or even the module) is finished.

Main Questions

1. (a) (i) Give an example (a_n) of an unbounded strictly increasing monotonic sequence of real numbers with the additional property that $a_n > 4$ for all $n \in \mathbb{N}$.
(ii) Give an example (b_n) of a bounded strictly decreasing monotonic sequence of real numbers with the additional property that $b_n < 2$ for all $n \in \mathbb{N}$.
(iii) Give an example (c_n) of a convergent sequence of real numbers with the additional property that even terms are greater than 3 and odd terms are less than 3.
(b) Give an example of a single sequence (x_n) that has each of the three sequences in part (a) as a subsequence.
2. Here are a number of possible distance functions on $\mathbb{R}$. Which, if any, is a metric on $\mathbb{R}$?
(a) $d_1(x, y) = \left| \sqrt{|x|} - \sqrt{|y|} \right|$,
(b) $d_2(x, y) = \left| \sqrt[3]{x} - \sqrt[3]{y} \right|$,
(c) $d_3(x, y) = \max(x, y) - \min(x, y)$,
(d) $d_4(x, y) = \min(x, y) - \max(x, y)$,
(e) $d_5(x, y) = e^x + e^y$,
(f) $d_6(x, y) = \begin{cases} 0 & \text{if } x = y, \\ e^x + e^y & \text{if } x \neq y. \end{cases}$
3. The expression $d(x, y) = |\log x - \log y|$ defines a distance function on the interval $(0, \infty)$. Is it a metric?
4. Show there are infinitely many metrics possible on a set with two or more members. How many metrics are possible on a singleton set?
5. In this question, we will consider the set of all horizontal line segments in the plane. A horizontal line segment in $(\mathbb{R}^2, d^{(2)})$ may be determined completely by its mid-point (a, b) and a number r that tells how wide the interval is. Thus

$$H(a, b; r) = \{(x, y) \in \mathbb{R}^2 : a - r \leq x \leq a + r \text{ and } y = b\}.$$

(Note: segments include their endpoints.)

Now let $\mathcal{H}$ be the set of such line segments in $(\mathbb{R}^2, d^{(2)})$, that is:

$$\mathcal{H} = \{H(a, b; r) : (a, b) \in \mathbb{R}^2, r \geq 0\}.$$

Define a distance function D on $\mathcal{H}$ by:

$$D(H(a, b; r), H(a', b'; r')) = d^{(2)}((a, a'), (b, b')) + |r - r'|$$

- (a) Sketch $H(-1, -1; 1)$, $H(1, 2; 3)$ and $H(0, 0; 0)$.
- (b) Calculate $D(H(-1, -1; 1), H(1, 2; 3))$.
- (c) Show that D defines a metric on $\mathcal{H}$.
- (d) Describe the unit sphere with centre $H(0, 0; 0)$.

6. Throughout this question we use the Euclidean metric for $\mathbb{R}$.

For each $n \in \mathbb{N}$, define a function $f_n : [0, \infty) \rightarrow \mathbb{R}$ by

$$f_n(x) = \begin{cases} \cos(\pi x) & \text{if } 2n - \frac{1}{2} \leq x \leq 2n + \frac{1}{2}, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Sketch, by hand or otherwise, the graphs of f_1, f_2 and f_3 .
 - (b) Determine whether the sequence (f_n) has a pointwise limit.
 - (c) Determine whether the sequence (f_n) converges uniformly on $[0, \infty)$.
7. Throughout this question we use the Euclidean metric for $\mathbb{R}$.
- For each $n \in \mathbb{N}$, define a function $f_n : [0, \infty) \rightarrow \mathbb{R}$ by

$$f_n(x) = \begin{cases} \frac{x(2n-x)}{n^2\sqrt{n}} & \text{if } 0 \leq x \leq n, \\ \frac{1}{\sqrt{n}} & \text{otherwise.} \end{cases}$$

- (a) Sketch, by hand or otherwise, the graphs of f_1, f_2 and f_3 .
 - (b) Determine whether the sequence (f_n) has a pointwise limit.
 - (c) Determine whether the sequence (f_n) converges uniformly on $[0, \infty)$.
8. For $n = 1, 2, 3, \dots$ define $f_n : [0, 1] \rightarrow [0, 1]$ by

$$f_n = \begin{cases} \frac{(n+1)x}{n} & \text{if } 0 \leq x \leq \frac{n}{n+1}, \\ (n+1)(1-x) & \text{if } \frac{n}{n+1} < x \leq 1. \end{cases}$$

Show that (f_n) has a pointwise limit on $[0, 1]$ and decide whether (f_n) is $d_{\max}$ convergent to the pointwise limit.

9. Show that the function $F : C[0, 1] \rightarrow C[0, 1]$ is $(d_{\max}, d_{\max})$ -continuous on $C[0, 1]$ where

$$F(f)(t) = \sin(t) + \int_0^t \cos(x)f(x) dx$$

for each $f \in C[0, 1]$ and each $t \in [0, 1]$.

10. For each $f \in C[0, 1]$ define $F : C[0, 1] \rightarrow \mathbb{R}$ by

$$F(f) = 2f(0) + 3 \int_0^1 sf(1-s)ds$$

Show that F is $(d_{\max}, d^{(1)})$ -continuous on $C[0, 1]$.

11. Let $A = \{(a, b) \in \mathbb{R}^2 : 2 < |a| + |b| \leq 4\}$ $B = \{(a, b) \in \mathbb{R}^2 : b < |a|\}$
and $C = A - B$.

- (a) Sketch A, B and C .
- (b) Is $(1, 1)$ a closure point of C with respect to the Euclidean metric $d^{(2)}$?
- (c) Write down in set notation the closure, interior and boundary of C with respect to $d^{(2)}$.
- (d) Write down, in set notation, the closure, interior and boundary of C for the ‘mixed’ metric d on $\mathbb{R}^2$ given by
 $d((x_1, x_2), (y_1, y_2)) = |x_1 - y_1| + d_0(x_2, y_2)$

12. (a) Can you find a single metric space (A, d) where:
- (i) The interval $(0, 1]$ is both d -open and d -closed,
 - (ii) The interval $[2, 3)$ is d -closed but not d -open,
 - (iii) The set $\{4\}$ is d -closed but not d -open,
 - (iv) The set $\{5\}$ is d -closed and d -open?
- (b) Is there a metric space (A, d) where, for some $a \in A$, $\{a\}$ is open but not closed?
13. Let $A = \{(a, b) \in \mathbb{R}^2 : |b| \geq a^2\}$, $B = \{(a, b) \in \mathbb{R}^2 : -1 < b \leq 1\}$, and $C = A \cap B$.
- (a) Sketch on separate diagrams each of A , B and C .
 - (b) Determine whether $(0, 0)$ is a $d^{(2)}$ -boundary point of C .
 - (c) Write down, in set notation, the closure, interior and boundary of C for:
 - (i) the Euclidean metric $d^{(2)}$,
 - (ii) the 'mixed' metric d on $\mathbb{R}^2$ given by $d((x_1, x_2), (y_1, y_2)) = |x_1 - y_1| + d_0(x_2, y_2)$,
 - (iii) the discrete metric d_0 .
14. B, C, D and E are subsets of $\mathbb{R}^2$ given by:
 $B = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 < 2\}$, $C = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 > |x_1|\}$,
 $D = \{(x_1, x_2) \in \mathbb{R}^2 : 2 < x_2 \leq 4\}$, $E = \{(x_1, x_2) \in \mathbb{R}^2 : -1 < x_1 \leq 1\}$
 and $A = (B \cap C) \cup (D \cap E)$.
- (a) Sketch the set A , following the conventions used in the module text for such sketches.
 - (b) Determine the d -interior, d -closure and d -boundary of C when d is:
 - (i) the Euclidean metric $d^{(2)}$,
 - (ii) the 'mixed' metric d given by $d((x_1, x_2), (y_1, y_2)) = d_0(x_1, y_1) + |x_2 - y_2|$,
 where (as usual) d_0 is the discrete metric on $\mathbb{R}$.
15. Throughout, for convenience, we use d for the Euclidean metric on $\mathbb{R}$.
 Further, we will use $\mathbb{P}$ to denote the set of prime numbers in $\mathbb{N}$.
 Let $S = \{x \in \mathbb{R} : px^2 = (n - 1)^2 \text{ for some } p \in \mathbb{P}, n \in \mathbb{N}\}$.
- (a) Show that S is countable.
 - (b) (i) Show that for each $y \in \mathbb{R}$ and each $r > 0$ there is $x \in S$ with $|x - y| < r$.
 (ii) Hence, or otherwise, show that S is d -dense in $\mathbb{R}$.

Extras

- (A) Show that the following defines a metric on the complex plane $\mathbb{C}$:

$$d(z_1, z_2) = \begin{cases} |z_1| + |z_2|, & \text{if } z_1 \neq z_2; \\ 0, & \text{if } z_1 = z_2. \end{cases}$$

This is sometimes known as the *British Rail*¹ metric.

Describe the open balls of radius 2, 4 and 6 centred at the point $2 + 2i$.

- (B) (a) Let G be a finite group and define a distance function on G by

$$d(a, b) = |(ab^{-1})| - 1, \text{ where } |g| \text{ denotes the order of } g \in G.$$

Show that d is not a metric on D_6 , but that d is a metric on $\mathbb{Z}_8$.

- (b) Now let G be a finite Abelian p -group. Determine whether d defines a metric on G .

- (C) Some letters in our alphabet may be constructed from line segments in the plane. For example the following union of three sets gives the letter “H”.

$$\{(0, b) \in \mathbb{R}^2 : 0 \leq b \leq 2\} \cup \{(1, b) \in \mathbb{R}^2 : 0 \leq b \leq 2\} \cup \{(a, 1) \in \mathbb{R}^2 : 0 \leq a \leq 1\}$$

Carrying on in this way determine the interior of HELL and the boundary of HELL with respect to:

- (a) the mixed metric d_1 given by $d_0(x_1, y_1) + |x_2 - y_2|$,
- (b) the mixed metric d_2 given by $|x_1 - y_1| + d_0(x_2, y_2)$.

- (D) (a) Let $\mathbb{Q}^* = \mathbb{Q} - \{0\}$ (that is, the rationals with 0 removed).

Is $\mathbb{Q}^*$ dense in $\mathbb{R}$ endowed with the Euclidean metric?

- (b) Let (X, d) be a metric space and let A be a subset of X that is d -dense in X . If we remove a point from A will the resulting set still be d -dense in X ?

¹The rather out-dated term refers to a belief that, in England, all rail travel had to go via London!